Guidelines for the Use of Peroral Endoscopic Myotomy (POEM) for the Treatment of Achalasia

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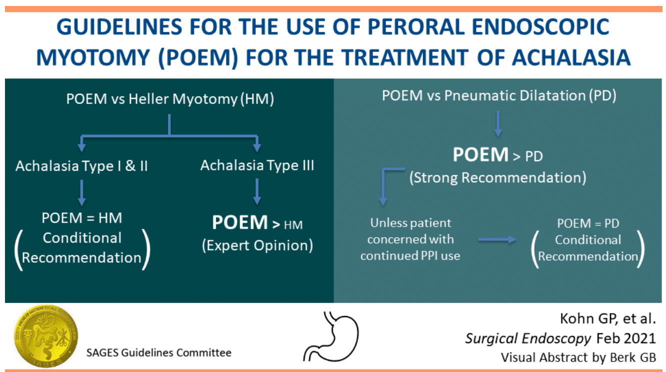

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Abstract

Background: Peroral Endoscopic Myotomy (POEM) is increasingly used as primary treatment for esophageal achalasia, in place of the options such as Heller myotomy (HM) and Pneumatic Dilatation (PD)

Objective: These evidence-based guidelines from the Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) intend to support clinicians, patients and others in decisions about the use of POEM for treatment of achalasia.

Results: The panel agreed on 4 recommendations for adults and children with achalasia.

Conclusions: Strong recommendation for the use of POEM over PD was issued unless the concern of continued postoperative PPI use remains a key decision-making concern to the patient. Conditional recommendations included the option of using either POEM or HM with fundoplication to treat achalasia, and favored POEM over HM for achalasia subtype III.

Keywords: Esophageal achalasia · POEM procedure · Heller myotomy · Pneumatic dilatation · Clinical practice guidelines

Executive Summary

Background

Peroral Endoscopic Myotomy (POEM) is increasingly used as primary treatment for esophageal achalasia, in place of the alternative endoscopic option of pneumatic dilatation of the lower esophageal sphincter, and in place of laparoscopic Heller myotomy with fundoplication. Many studies have compared POEM to Heller myotomy (HM), and some have compared it to pneumatic dilation (PD), with much heterogeneity in outcomes and findings. Guidelines are required to guide the endoscopist (surgeon or gastroenterologist) in the choice of procedure based on a systematic synthesis of best available evidence.

Interpretation of strong and conditional recommendations

The strength of these guideline recommendations is either “strong” or “conditional” as per the GRADE approach [1]. The words “the guideline panel recommends” are used for strong recommendations, and “the guideline panel suggests” for conditional recommendations, according to the GRADE approach. The strength of the recommendation is to be considered according to the following, an adaptation of standard interpretation[2]:

Strong recommendation

- For patients: almost all informed patients in this situation would want the recommended course of action.
- For clinicians: almost all clinicians should follow the recommended course of action guided by the best available evidence suggesting desirable effects clearly outweigh undesirable effects
- For policy makers: the recommendation can be adopted as policy in most situations. Adherence to this recommendation according to the guideline could be used as a quality criterion or performance indicator.

Conditional recommendation

- For patients: the majority of individuals in this situation would want the suggested course of action, but a substantial proportion may not. Decision aids may be useful in helping patients to make decisions consistent with their individual risks, values, and preferences.
- For clinicians: different choices will be appropriate for individual patients, and clinicians must help each patient arrive at a management decision consistent with the patient's values and preferences. Decision aids may be useful in helping individuals to make decisions consistent with their individual risks, values, and preferences.
- For policy makers: policy making should support the course of action as a valid treatment and exact indications will require debate and involvement of various stakeholders. Performance measures about the suggested course of action should focus on whether an appropriate decision-making process is duly documented.

How to use these guidelines

These guidelines are primarily intended to help endoscopists (surgeons and gastroenterologists) make decisions about management of their patients. Other purposes are to educate, inform policy and advocacy and to define future research needs. Guidelines are applicable to all physicians facing patient management uncertainties addressed herein without regard to specialty, training or interests. Due to the complexity of the healthcare environment, these guidelines are intended to indicate the preferred, but not necessarily the only, acceptable approach to management. Guidelines are intended to be flexible depending on individual circumstances. Given the wide range of specifics in any healthcare problem, the surgeon must always choose the course best suited to the individual patient and the variables in existence at the moment of decision. These Guidelines can also be used by patients as a basis of discussion with their treating surgeon.

Key questions addressed by these guidelines

1: Should peroral endoscopic myotomy (POEM) vs. Heller myotomy (HM) be used for achalasia in adults and children?

2: Should peroral endoscopic myotomy (POEM) vs. Pneumatic Dilatation (PD) be used for achalasia in adults and children?

Recommendations

Should Peroral Endoscopic Myotomy (POEM) vs. Heller myotomy (HM) be used for achalasia in adults and children?

1. **The Guideline panel suggests that adult and pediatric patients with type I and II achalasia may be treated with either POEM or laparoscopic Heller myotomy based on surgeon and patient's shared decision-making (conditional recommendation, very low certainty evidence).**
2. **Based on their collective experience, the panel suggests POEM over laparoscopic Heller myotomy for type III adult or pediatric achalasia (expert opinion).**

Should Peroral Endoscopic Myotomy (POEM) vs. Pneumatic Dilatation (PD) be used for achalasia in adults and children?

1. **The Guideline panel recommends peroral endoscopic myotomy over pneumatic dilatation in patients with achalasia (strong recommendation, moderate certainty evidence).**
2. **For the subgroup of patients who are particularly concerned about the continued use of PPI post-operatively , the panel suggests that either POEM or pneumatic dilatation can be used based on joint patient and surgeon decision-making (conditional recommendation, very low certainty evidence).**

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Introduction

Aim of these guidelines and specific objectives

The purpose of these guidelines is to provide evidence-based recommendations regarding the utility of Peroral Endoscopic Myotomy (POEM) in the management of esophageal achalasia. Taking a surgeon-patient perspective, the key target audience includes patients, surgeons, gastroenterologists and endoscopists. Policy makers and insurance providers involved in delivering local, national and international healthcare services aimed at the treatment of achalasia or involved in evaluating direct and indirect benefits, harms and costs related to the various procedures used to treat the disease may also consider these recommendations in their deliberations.

Description of the health problems

Achalasia is an esophageal motility disorder characterized by aperistalsis of the esophageal body and the absence of relaxation of the lower esophageal sphincter. Passage of food into the stomach is impaired, and dysphagia is frequently experienced, often with chest pain. Regurgitation of undigested food may occur, and respiratory symptoms may result in cough, aspiration and possibly pneumonia [3]. The incidence of the disorder is approximately 2.5 per 100,000 people per year in adults [4,5], with incidence increasing with age[6]. Achalasia occurs as an effect of the destruction of enteric neurons controlling lower esophageal sphincter (LES) relaxation and esophageal body musculature by an unknown cause, most likely inflammatory [7].

High-resolution esophageal manometry is used to diagnose achalasia and to subdivide the disorder into three distinct subtypes; type I (classical achalasia with no esophageal pressurization), type II (achalasia with esophageal compression) and type III (achalasia with esophageal spasm)[8].

With the absence of any known therapies able to restore these enteric neurons and reestablish esophageal body motility, treatments are directed towards disruption of the lower esophageal sphincter and decreasing resistance to esophageal emptying. Pneumatic dilation (PD), laparoscopic Heller myotomy (LHM) and peroral endoscopic myotomy (POEM) have all been used to lower LES pressure, and each has its benefits and disadvantages. These guidelines provide recommendations regarding the use of POEM in the treatment of achalasia.

Methods

The creation of this guideline followed SAGES policies established to meet recommendations for trustworthy guidelines made by the Institute of Medicine [9]. This process included the panel formation, organizational approval and review, and plans for updating.

A systematic review of the literature was performed by the SAGES Guidelines Committee to inform the panel's evidence-based deliberations and recommendations [10]. Reported according to the Preferred reporting items for systematic reviews and meta-analyses (PRISMA) checklist, the systematic review will be published separately.

The guideline panel used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) Evidence to Decisions (EtD) approach to deliberate and formulate recommendations [11,12] .

Reporting of this guideline was structured as per the Essential Reporting Items for Practice Guidelines in Healthcare (RIGHT) checklist [13]

Guideline Panel Organization

International experts on POEM were invited to participate in the SAGES Guideline Panel. All panel members were experienced endoscopists with training in either surgery or gastroenterology and submitted disclosures on potential conflicts of interest. The Chair of the panel (G.K.) declared no conflicts of interest and assessed the conflicts of the panelists as to be not relevant and not likely to influence the direction of strength of the recommendations. Additional voting members included patient advocates (C.R. and J.C.) who were the president and member of the International Foundation for Gastrointestinal Disorders (IFFGD), respectively. A methodologist with extensive guideline development experience (M.A) and the SAGES Guidelines Committee Fellow (R.D.) participated on the panel as non-voting members and facilitated appraisal of the evidence and formulation of the recommendations. A full list of all contributors to the guideline development is provided in Appendix A .

The panel reviewed the literature presented by the SAGES systematic review working group and met for half-day video conferences over the course of three online sessions to create recommendations. The panel reviewed evidence tables and voted on components of Evidence to Decision tables to reach final recommendations. Both the evidence tables and evidence to decision tables were compiled using GradePro[14].

Guideline funding & Declaration and management of competing interests

Funding for the methodologist and librarians came from SAGES as did half the salary of the Guidelines Committee Fellow. No industry support was used to create this guideline, nor was any industry input used for any stage of the development, dissemination, or implementation of this guideline. Standard disclosure forms were completed by all guideline contributors to evaluate for potential conflict of interest. Evaluation of these conflict was made by the panel Chair, and no potential conflicts were deemed to have affected the decision. A full list of declarations can be found in Appendix B.

Selection of questions and outcomes of interest

A systematic review working group was formed from the SAGES Guidelines Committee and tasked to perform a literature search on POEM. Heller myotomy and pneumatic dilation were identified as comparators of interest, and key questions were formulated according to the patient-intervention-comparator outcome (PICO) format. Outcomes important for decision-making were determined *a priori* to be “critical” or “important” for patients. These outcomes centered on efficacy, safety, and side effects associated with POEM, Heller Myotomy, and

Pneumatic Dilation. These outcomes including dysphagia rates, quality of life, patient satisfaction, reoperation, perforation, pain scores, reflux symptoms, bloating and flatulence. Cost and length of stay were also included.

Evidence Appraisal

Results from the SAGES systematic review and meta-analysis were uploaded to GradePro to facilitate evidence appraisal and panel decision-making. Details for the original systematic review will be published separately in Surgical Endoscopy[10]. Evidence that directly compared POEM to either Heller myotomy or PD was used in this guideline. Indirect evidence was not deemed necessary and omitted in favor of direct evidence. When data for a particular outcome was reported in both a higher certainty RCT and a lower certainty observational study, the observational data was omitted if the direction of the effect was concordant. If some observational studies had no data unique to that of an RCT, and if the data was concordant in direction, then the observational study was omitted.

To judge the certainty of available evidence, risk of bias was assessed across included studies for each outcome, as was inconsistency, indirectness and imprecision. Because low numbers of studies precluded appropriate use of common measures of publication bias such as Egger's and Begg's test or funnel plots, selective reporting within studies was considered during individual risk of bias assessment instead of across studies with reporting bias. Together, these considerations contributed to an overall certainty for each outcome. Any time a study was downgraded in certainty, an explanation was given in a footnote on the evidence table.

Development of clinical recommendations

Using the GRADE approach, the panel considered the evidence in making judgments across the criteria in the EtD framework. These criteria include judgments about the magnitude of the desirable and undesirable effects, the overall certainty of the evidence, the potential variation in values of key stakeholders, the balance of these effects, and the acceptability and feasibility of the option favored by the balance of effects. Cost-effectiveness was not considered as the guideline took the patient-surgeon rather than a societal perspective, but available cost studies were listed as additional considerations for feasibility and acceptability.

After due deliberations, the panel voted to finalize judgement on each component of the EtD table. If judgments differed considerably among panelists, additional discussion was held with panel members advocating alternative response options. Dissenting views were transparently documented. Re-voting was then employed to reach majority consensus on each EtD criterion specific judgment. Final guideline recommendations were based on ≥ 70% agreement.

The panel addressed subgroups such as pediatric patients and achalasia subtypes during a discussion for the justification for their recommendation. These considerations are specified for each key question below. The evidence, the additional considerations, and final judgements for each step in the decision-making process are presented in the EtD tables in Appendix C and summarized in the recommendations that follow.

Values and preferences

Because expert opinion suggested absence of literature investigating patient values and preferences related to this topic, values and references literature was not specifically searched. Practicing panel members and patient advocates represented patient values and preferences to inform these guidelines.

Guideline Document review

All panel members reviewed and provided comments and edits on the guideline. The final draft was distributed first to the SAGES Guidelines Committee for comments and approval and then to the SAGES Board for comments and approval. The document approved by the SAGES Board was then published online on the SAGES website (https://www.sages.org/publications/guidelines/) for public comment for 1 month. After this public comment period, the final version of this guideline was submitted for publication.

Recommendations

Key question 1: *Should peroral endoscopic myotomy (POEM) vs. Heller myotomy (HM) be used for achalasia in adults and children?*

1. **The Guideline panel suggests that adult and pediatric patients with type I and II achalasia may be treated with either POEM or laparoscopic Heller myotomy based on surgeon and patient's shared decision-making (conditional recommendation, very low certainty evidence).**
2. **Based on their collective experience, the panel suggests POEM over laparoscopic Heller myotomy for type III adult or pediatric achalasia (expert opinion).**

Summary of the evidence

From twenty-one included studies in the systematic review, one recent randomized control trial with low risk of bias comparing POEM versus laparoscopic Heller myotomy with Dor fundoplication, and fourteen high risk of bias observational studies comparing POEM versus laparoscopic Heller myotomy with predominantly Dor or Toupet fundoplication were used to inform the panel's decision.

Benefits

POEM had similar effectiveness as Laparoscopic Heller myotomy with fundoplication (LHM) with regards to:

- Success/symptom resolution, defined by Eckardt Score ≤3 (1 RCT of 221 participants, RR 1.02 CI [0.90-1.15], absolute difference 1.6% more CI [8.2 fewer to 12.2 more]);
- Postoperative LES relaxation pressures (1 RCT of 148 participants, mean difference 0.75 mmHg lower CI[2.26 lower to 0.76 higher]); *and*
- GI quality of life improvement at 2 years (1 RCT of 202 participants, mean difference 0.14 higher CI [4.01 lower to 4.28 higher]).

The combined magnitude of these effects was thought to be trivial by the panelists.

POEM was better, to a level determined to be of clinical significance by the panel, in the outcomes of:

- Postoperative pain (3 observational studies of 269 total participants, RR 0.88 CI [0.60-1.29], 2.6% fewer CI [8.6% fewer to 6.2% more];
- Serious adverse events (1 RCT of 221 participants, RR 0.36 CI [0.10-1.34], 4.7% fewer CI [6.6% fewer to 2.5% more]; *and*
- Return to the operating room for postoperative complications (9 observational studies of 618 participants, RR 0.79 CI [0.28-2.22], 0.5% fewer CI [1.7% fewer to 2.9% more]).

Harms and burden

The undesirable effect of reflux esophagitis (of Los Angeles Grades B, C or D) was seen more frequently at 2 years in the POEM group as compared to the Heller myotomy with fundoplication group (1 RCT of 165 participants, RR 1.79 CI[0.90-3.59], 10.1% more CI[1.3% fewer to 33.2% more]. However, when only more severe esophagitis of Los Angeles grades C or D were evaluated, the effect favors POEM with RR 0.72 CI [0.20, 2.58], though still without a statistically significant difference.

Cost

Four observational studies provided information on cost (in USD) for both POEM and LHM, though without any measure of variance for a meta-analysis to be performed. Despite great variation in components of cost, three of four studies stated POEM was more expensive, but the estimated cost difference was small ($3301 cheaper to $1026 more expensive, for the index admission).

Certainty in the evidence of effects

The certainty in these effects was rated as low or very low owing to the risk of bias and imprecision of the estimates. The overall certainty of evidence was deemed very low. (see evidence profile in the EtD framework, Appendix C).

Decision criteria and additional considerations

For this judgement the panel considered the value for decision-making that informed patients would place on the main outcomes based on their experience and the available evidence. In addition, the expert panel's opinion was influenced by the patient participants' expressed values. Although the panel felt that there was probably no important variation, they agreed that given limited evidence on patient values, variation was likely. Additionally, a minority of the panel, notably a patient member of the panel, felt there may "possibly" be important variability.

Outcomes of "critical" importance for decision-making were determined to be: success and symptom resolution, quality of life improvement, serious adverse events, return to the operating room for complications, and reflux esophagitis at 2 years. Outcomes deemed "important" were thought to be of lesser significance to a well-informed patient and included postoperative LES relaxation pressures and early postoperative pain.

Based on expert opinion, the panel believed that POEM had additional desirable effects not assessed by the reviewed literature including lack of incisional hernia risk, lower wound infection rate, and lack of post-fundoplication side effects such as bloating, flatulence, and inability to belch or vomit.

The only undesirable effect (reflux esophagitis on EGD) was considered to be a moderate effect as a short-term outcome but was downgraded to a small effect as the biggest difference between procedures was seen for Grade A esophagitis and not for the more clinically relevant grades B-D esophagitis. The panel felt longer-term comparative data is needed to assess reflux outcome differences between the procedures in the long term, including esophagitis and its sequalae.

The systematic review that preceded this guideline revealed that subtype of achalasia was often either not mentioned, or reported outcomes were not stratified by subtype, precluding subgroup analyses on POEM versus LHM for each achalasia subtype.[10] In those studies which reported distribution of achalasia subtype, types I or II were usually predominant (averages of 70%-100% sub types I or II in 16 of 18 studies) with only one study in the systematic review reporting predominantly Type III patients. The panel's recommendations presented here thus apply best to achalasia subtypes I and II. Limited evidence is available on POEM versus LHM in type III achalasia patients to make an evidence-based recommendation. Based on expert opinion, however, POEM appears to perform better in type III patients as a longer myotomy can be performed with that approach. Therefore, for type III achalasia patients, this panel suggested POEM for its higher efficacy but deemed both POEM and LHM as safe choices.

Conclusion

Both desirable and undesirable anticipated effects for POEM were judged to be small. Some panel members expressed an opinion that the balance of desirable and undesirable effects favored the intervention (POEM) but after further deliberation among panel members, the panel unanimously agreed that the balance of effects did not favor one over the other.

The panel suggests that adult patients with achalasia subtypes 1 and 2 may be treated with <u>either</u> POEM or laparoscopic Heller myotomy with fundoplication based on surgeon and patient's shared decision-making. Given the lack of contradictory data in children, this recommendation may also be generalized to the pediatric population.

No evidence-based recommendation can be made for patients with achalasia subtype III. However, as expert opinion, the panel favored POEM over laparoscopic Heller myotomy for type III adult or pediatric achalasia.

Both LHM and POEM are already established procedures. The panel felt there would be clear acceptance for the recommendation to perform either procedure. While some specific practitioners who are unfamiliar or inexperienced with POEM may not find a recommendation to be feasible for POEM in preference to LHM, the recommendation for either POEM or LHM was considered generally feasible as currently there are numerous groups that have accumulated experience with and regularly perform POEM.

Panel recommendations for future research for this recommendation

The panel made multiple suggestions for future research priorities:

- More research is needed on achalasia subtype III-specific outcomes after POEM vs. Heller. This can be achieved either with type III-only study populations, or studies with sample size large enough to perform adequately powered subgroup analysis based on achalasia subtype.
- More research is needed on pediatric populations. This can be achieved either with pediatric only studies, or studies with sample size large enough to perform adequately powered subgroup analysis based on pediatric versus adult populations.
- Longer term results are needed for all outcomes given the chronic nature of achalasia. Surveillance and follow-up past 10 years is needed, especially by high quality comparative studies.
- Future studies should include better measures to determine the presence of dysphagia, rather than Eckardt score which tends to be not very specific. More accurate, objective alternatives include manometry and timed barium swallow studies.

- More research is needed into outcomes of POEM versus Heller myotomy that relate specifically to the fundoplication component of a Heller myotomy.
- Research should be performed to establish whether there is a correlation between post-POEM LES pressure and post-POEM outcomes. Such data exist for Heller myotomy but not for POEM and the panel felt that it is not appropriate to apply the evidence from Heller to POEM as there may be substantial differences between the procedures.
- Current evidence suggests POEM leads to greater postoperative reflux, at least in the first 2 years post procedure. However, there is no research on the role, patient acceptance, and efficacy of PPI use after POEM for this undesirable outcome. The panel recommends further investigation of strategies to address undesirable effects for both POEM and Heller myotomy and their relative efficacy for both interventions.

Key question 2:
Should peroral endoscopic myotomy (POEM) vs. Pneumatic Dilatation (PD) be used for achalasia in adults and children?

1. **The Guideline panel recommends peroral endoscopic myotomy over pneumatic dilatation in patients with achalasia (strong recommendation, moderate certainty evidence)**
2. **For the subgroup of patients who are particularly concerned about the continued use of PPI post-operatively , the panel suggests that either POEM or pneumatic dilatation can be used based on joint patient and surgeon decision-making (conditional recommendation, very low certainty evidence).**

Summary of the evidence

From evidence from eight studies included in the systematic review, one recent randomized control trial on POEM versus pneumatic dilatation, and five predominantly high risk of bias observational studies on POEM versus PD, informed the panel’s decision.

Benefits

POEM was superior to Pneumatic Dilation in nearly all desirable effects:

- Success/symptom resolution by Eckardt Score, at 2 years (1 RCT, 126 participants, RR 1.71 CI [1.34-2.17], 38.5% more CI [18.3% more – 63.1% more]);
- Short-term Eckardt scores at 0-6 months (6 observational studies, 471 participants, mean difference 0.7 lower CI [1.25 lower to 0.16 lower]);
- Requirement for retreatment for failure of symptom relief (1 RCT, 126 participants, RR 0.19 CI [0.08-0.47], 33.4% fewer CI [38% fewer – 21.9% fewer];
- Treatment-related serious adverse events (1 RCT, 126 participants, RR 0.33 CI [0.01-8.21], 1.1% fewer CI [1.6% fewer – 11.4% more]); *and*

- Achalasia disease-specific quality of life scores (1 RCT, 92 participants, mean difference: 0, CI [3 lower – 3 higher])

The panel unanimously agreed that the desirable effects of POEM compared to PD were large.

Harms and burden

POEM resulted in more:

- PPI use at 2 years (1 RCT, 92 participants, RR 2.01 CI [0.97-4.16], 20.8% more CI [0.6% fewer to 65.1% more]) *and*
- More patient-reported short-term reflux symptoms (3 observational studies with a total of 164 participants, RR 2.67 CI [1.02-7.0], 9.8% more CI [0.1% more – 35.3% more])

Objective evidence of GERD includes esophagitis. However, esophagitis data was limited, and an absolute effect could not be calculated on GradePro given the lack of esophagitis events in the control (PD) group. No observational evidence was available to strengthen this evidence. Thus, though a critical outcome, esophagitis was removed due to inadequate evidence. The panel believed that POEM does likely pose an increased risk for esophagitis, but the degree of this effect cannot be gauged by the available evidence.

These two harm effects listed are proxy markers for postoperative GERD. The main undesirable effect in the available research was PPI use. The panel believed the degree of undesirable effect ultimately varies based on the importance placed by patients on PPI therapy post-intervention. The panelists varied in whether post-intervention PPI should even be included as a decision-making outcome, as PPI use often does not correlate with objective measures of reflux. As such, a significant proportion of informed patients would likely consider PPI use as low importance for decision-making and the remaining undesirable outcome of short-term reflux symptoms could be judged as having either a trivial or small effect.

Notwithstanding, the panel acknowledged the subgroup of patients for whom PPI-use would be an important or even critical decision-making outcome, particularly patients who opt for the procedure because of their concerns about long-term PPI use. For this subgroup of patients, PPI use would be an important outcome for decision-making, with moderate or even large magnitude of the observed undesirable effect. Unfortunately, defined criteria for the use of PPIs is absent in the studies, and the appropriateness of PPI use in these patients is therefore unknown.

Decision criteria and additional considerations

The panelists agreed there would unlikely be any variability in how patients value the main outcomes related to efficacy or safety. Based on experience with this patient population, the panel was certain that patients value dysphagia and procedure-related adverse events as critical decision-making outcomes.

However, there was extensive debate on the value of post-intervention PPI use for patient decision-making, suggesting there possibly may be variability in how this outcome is valued by patients. Given the large magnitude of this outcome’s effect, any such variability in values would be important to decision-making.

Conclusions

The panelists universally agreed that the evidence provided favored POEM over PD (10% probably favors, 90% favors).

Esophagitis was still considered a critical outcome that likely favored PD based on expert opinion. However, the panelists agreed the unknown degree of effect from esophagitis was not enough to outweigh the large effect from evidence supporting POEM.

Panelists agreed that this intervention was acceptable or probably acceptable, with a simple majority (60%) favoring acceptable.

Economic considerations for the patient were included in feasibility. There were a wide range of opinions on feasibility within the panel. Concerns for feasibility included the availability of endoscopists trained to perform POEM and the potential increased out-of-pocket costs for POEM in certain countries and health systems.

Due to the variability and inequity in insurance markets, even within a single country such as the USA, the panel was unable to analyze the feasibility based on insurance coverage. However, the panel agreed that increased coverage by insurance would improve feasibility of performing the recommended procedure.

The panel also considered the feasibility of the pneumatic dilatation and felt that accessibility of an endoscopist trained to perform esophageal dilatation with an achalasia balloon was also limited in certain regions, and that concerns with feasibility were similar for POEM and PD.

Today, training is available for endoscopists for POEM. Increased teaching and training will be needed to improve the accessibility to patients and thus feasibility. In regions with poor insurance coverage for POEM and high out-of-pocket expenses, improved insurance coverage would improve feasibility.

Panel recommendations for future research for this recommendation

The panel makes multiple suggestions for future research priorities:

- More research on achalasia treatment in pediatric populations. This can be achieved either with pediatric only studies, or studies with sample size large enough to perform adequately powered subgroup analysis based on pediatric versus adult population.
- More research on achalasia subtype III patients.
- Studies with long term follow-up measures to determine the incidence and severity of esophagitis after POEM, as well as incidence of sequelae of esophagitis.
- Further research on the role, patient acceptance, and efficacy of PPI use after POEM

What are others saying, and what is new in these SAGES guidelines?

Other recent guidelines exist to guide treatment of achalasia [7, 15-17]. Concerns are expressed about the incidence of GERD after POEM [15], though esophagitis or GERD symptoms are not evaluated by severity or extent, and either treatment option is usually suggested, similar to the recommendation of this guideline [7, 16, 17]. As recommended in this guideline, POEM is thought to be the treatment of choice for achalasia subtype III [16, 17]. Other guidelines have also suggested that the management of achalasia in the pediatric population can follow recommendations made for the adult population [15].

The main difference between this guideline and others is this guidelines' recommendation for POEM over pneumatic dilation, whereas other guidelines have determined that PD is a reasonable first-line treatment of achalasia [16].

Implementation and revision of these guidelines

Implementation

The panel believes that it is feasible to successfully implement these recommendations into local practice and that the recommendations will be accepted by stakeholders. Regarding the suggestion that either POEM or Heller may be used in adult patients with achalasia, this allows for both procedures to continue in the community and third party payers are urged to reimburse for either procedure. Where practitioners are unfamiliar with one or the other technique, training courses, proctors and supervisors are widely available to most practitioners.

The number of practitioners experienced in the use of balloon dilatation with achalasia balloons (rather than simple through-the-scope balloons) is not large and is thought to be decreasing with time. Therefore, the pneumatic dilation procedure is being performed less frequently and again, since training opportunities are widespread, uptake of POEM should be feasible in most jurisdictions.

Updating these guidelines

After publication of these Guidelines, SAGES will maintain them through surveillance for new evidence, with repeat literature searches performed at regular intervals. A formal update will be undertaken, if determined that there have been interim major changes in the evidence base, in 3 years post-publication.

Limitations of these guidelines

The limitations of these guidelines are inherent to the very low certainty of the evidence we identified for Key Question 1. This was a lesser problem with the moderate certainty identified for Key Question 2. The current data, particularly for Key Question 1, are limited by the short-term follow-up period (up to 2 years). The panel considered these data as a proxy for long-term outcomes (5-year, 10-year, and 15-year follow-up). This is the best available proxy, based on the panel's opinion. However, the panel expressed an interest in following up on any longer-term data in the future, to abolish the need for this proxy.

Research priorities summarized

More research is needed on Type III achalasia specific outcomes after POEM vs. Heller. This can be achieved either with type III-only study populations, or studies with sample size large enough to perform adequately powered subgroup analysis based on achalasia subtype.

More research is needed on pediatric populations. This can be achieved either with pediatric only studies, or studies with sample size large enough to perform adequately powered subgroup analysis based on pediatric versus adult population.

Longer term results are needed for all outcomes given the chronic nature of achalasia. Surveillance and follow-up past 10 years is needed, especially by high quality comparative studies.

Future studies should include better measures to determine the presence of dysphagia than Eckardt score which tends to be not very specific. More accurate, objective alternatives include manometry and timed barium swallow studies.

More research is needed into outcomes of POEM versus Heller myotomy that relate specifically to the fundoplication component of a Heller myotomy.

Research should be performed to establish whether there is a correlation between post-POEM LES pressure and post-POEM outcomes. Such data exist for Heller myotomy but not for POEM and the panel felt that it is not appropriate to apply the evidence from Heller to POEM as there may be substantial differences.

While the current evidence suggests POEM leads to greater postoperative reflux, at least in the first 2 years post procedure, there is no research on the role, patient acceptance and efficacy of PPI use after POEM for this undesirable outcome. The panel recommends further investigation of strategies to address undesirable effects for both POEM and Heller myotomy and their relative efficacy for both interventions.

Studies with long term follow-up measures to determine the incidence and severity of esophagitis after POEM, as well as incidence of sequelae of esophagitis.

Further research on the role, patient acceptance, and efficacy of PPI use after POEM.

Acknowledgements

The authors thank Sarah Colon (SAGES) for administrative support.

Authorship:

GPK was the panel Chair, wrote the first draft of the manuscript and revised the manuscript based on author's suggestions; RCD was the panel co-Chair, contributed to drafting and critical revisions of the manuscript and contributed to further drafts, and moderated the panel sessions, and checked the manuscript accuracy; MA provided methodological support; Guideline panel members (GPK, JC, CD, LL, JM, DM, CR, PS, LS, RW, AP, DS) participated in the creation of the EtD tables, critically reviewed the manuscript and provided suggestions for improvement; All authors approved the content.

References

1. Andrews JC, Schünemann HJ, Oxman AD, Pottie K, Meerpohl JJ, Coello PA, Rind D, Montori VM, Brito JP, Norris S, Elbarbary M, Post P, Nasser M, Shukla V, Jaeschke R, Brozek J, Djulbegovic B, Guyatt G (2013) GRADE guidelines: 15. Going from evidence to recommendation-determinants of a recommendation's direction and strength. J Clin Epidemiol 66:726-735
2. Andrews J, Guyatt G, Oxman AD, Alderson P, Dahm P, Falck-Ytter Y, Nasser M, Meerpohl J, Post PN, Kunz R, Brozek J, Vist G, Rind D, Akl EA, Schünemann HJ (2013) GRADE guidelines: 14. Going from evidence to recommendations: the significance and presentation of recommendations. J Clin Epidemiol 66:719-725
3. Boeckxstaens GE, Zaninotto G, Richter JE (2014) Achalasia. Lancet 383:83-93
4. Duffield JA, Hamer PW, Heddle R, Holloway RH, Myers JC, Thompson SK (2017) Incidence of achalasia in South Australia based on esophageal manometry findings. Clin Gastroenterol Hepatol 15:360-365
5. Samo S, Carlson DA, Gregory DL, Gawel SH, Pandolfino JE, Kahrilas PJ (2017) Incidence and prevalence of achalasia in Central Chicago, 2004-2014, since the widespread use of high-resolution manometry. Clin Gastroenterol Hepatol 15:366-373

6. Farrukh A, DeCaestecker J, Mayberry JF (2008) An epidemiological study of achalasia among the South Asian population of Leicester, 1986-2005. Dysphagia 23:161-164
7. Oude Nijhuis RAB, Zaninotto G, Roman S, Boeckxstaens GE, Fockens P, Langendam MW, Plumb AA, Smout A, Targarona EM, Trukhmanov AS, Weusten B, Bredenoord AJ (2020) European guidelines on achalasia: United European Gastroenterology and European Society of Neurogastroenterology and Motility recommendations. United European Gastroenterol J 8:13-33
8. Kahrilas PJ, Bredenoord AJ, Fox M, Gyawali CP, Roman S, Smout AJ, Pandolfino JE, International High Resolution Manometry Working Group (2015) The Chicago Classification of esophageal motility disorders, v3.0. Neurogastroenterol Motil 27:160-174
9. Institute of Medicine (U.S.) Committee on Standards for Developing Trustworthy Clinical Practice Guidelines, Graham R (2011) Clinical practice guidelines we can trust, National Academies Press, Washington, DC
10. Dirks RC, Kohn GP, Slater B, Whiteside J, Rodriguez NA, Docimo S, Pryor A, Stefanidis D; SAGES guidelines committee. Is peroral endoscopic myotomy (POEM) more effective than pneumatic dilation and Heller myotomy? A systematic review and meta-analysis. Surg Endosc. 2021 May;35(5):1949-1962.
11. Alonso-Coello P, Schünemann HJ, Moberg J, Brignardello-Petersen R, Akl EA, Davoli M, Treweek S, Mustafa RA, Rada G, Rosenbaum S, Morelli A, Guyatt GH, Oxman AD (2016) GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well informed healthcare choices. 1: Introduction. BMJ 353:i2016
12. Alonso-Coello P, Oxman AD, Moberg J, Brignardello-Petersen R, Akl EA, Davoli M, Treweek S, Mustafa RA, Vandvik PO, Meerpohl J, Guyatt GH, Schünemann HJ; GRADE Working Group (2016) GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well informed healthcare choices. 2: Clinical practice guidelines. BMJ 353:i2089
13. Chen Y, Yang K, Marusic A, Qaseem A, Meerpohl JJ, Flottorp S, Akl EA, Schunemann HJ, Chan ES, Falck-Ytter Y, Ahmed F, Barber S, Chen C, Zhang M, Xu B, Tian J, Song F, Shang H, Tang K, Wang Q, Norris SL, RIGHT Working Group (2017) A reporting tool for practice guidelines in health care: The RIGHT Statement. Ann Intern Med 166:128-132
14. McMaster University (2015) GRADEpro GDT: GRADEpro Guideline Development Tool. [Software]. (developed by Evidence Prime, Inc.). Available from http://gradepro.org

15. Zaninotto G, Bennett C, Boeckxstaens G, Costantini M, Ferguson MK, Pandolfino JE, Patti MG, Ribeiro U, Jr., Richter J, Swanstrom L, Tack J, Triadafilopoulos G, Markar SR, Salvador R, Faccio L, Andreollo NA, Cecconello I, Costamagna G, da Rocha JRM, Hungness ES, Fisichella PM, Fuchs KH, Gockel I, Gurski R, Gyawali CP, Herbella FAM, Holloway RH, Hongo M, Jobe BA, Kahrilas PJ, Katzka DA, Dua KS, Liu D, Moonen A, Nasi A, Pasricha PJ, Penagini R, Perretta S, Sallum RAA, Sarnelli G, Savarino E, Schlottmann F, Sifrim D, Soper N, Tatum RP, Vaezi MF, van Herwaarden-Lindeboom M, Vanuytsel T, Vela MF, Watson DI, Zerbib F, Gittens S, Pontillo C, Vermigli S, Inama D, Low DE (2018) The 2018 ISDE achalasia guidelines. Dis Esophagus 31 DOI: 10.1093/dote/doy071
16. Jung HK, Hong SJ, Lee OY, Pandolfino J, Park H, Miwa H, Ghoshal UC, Mahadeva S, Oshima T, Chen M, Chua ASB, Cho YK, Lee TH, Min YW, Park CH, Kwon JG, Park MI, Jung K, Park JK, Jung KW, Lim HC, Jung DH, Kim DH, Lim CH, Moon HS, Park JH, Choi SC, Suzuki H, Patcharatrakul T, Wu JCY, Lee KJ, Tanaka S, Siah KTH, Park KS, Kim SE, Korean Society of Neurogastroenterology and Motility (2020) 2019 Seoul Consensus on Esophageal Achalasia Guidelines. J Neurogastroenterol Motil 26:180-203
17. Khashab MA, Vela MF, Thosani N, Agrawal D, Buxbaum JL, Abbas Fehmi SM, Fishman DS, Gurudu SR, Jamil LH, Jue TL, Kannadath BS, Law JK, Lee JK, Naveed M, Qumseya BJ, Sawhney MS, Yang J, Wani S (2020) ASGE guideline on the management of achalasia. Gastrointest Endosc 91:213-227 e6

Abbreviations & Acronyms

POEM = Peroral Endoscopic Myotomy
HM = Heller myotomy
LHM = Laparoscopic Heller Myotomy
PD = Pneumatic dilation
GERD = Gastroesophageal reflux disease
RCT = Randomized Controlled Trial
RR = Risk ratio
CI = Confidence Interval
EtD = Evidence to Decision
PPI = Proton pump inhibitor

Appendices

A. List all contributors, their roles and affiliations

B. Full declarations of Conflicts of interest

C. Evidence to Decision Tables

Appendix A – List of contributors

Panelists

Surgeons: Geoffrey P Kohn: Monash University, VIC, Australia; Rebecca C Dirks: Indiana University, IN, USA; Christy M Dunst: The Oregon Clinic, OR, USA; Lars Lundell: Karolinska University Hospital, Sweden; Jeffrey Marks: University Hospitals Cleveland Medical Center, OH, USA; Daniele Molena: Memorial Sloan Kettering Cancer Center, NY, USA; Lee Swanstrom: Institut de Chirugie, Strasbourg, France; Aurora D Pryor: Stony Brook University, NY, USA; Dimitrios Stefanidis: Indiana University, IN, USA

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Appendix B – Full declarations of Conflicts of interest

Conflicts of Interest disclosures

All conflicts of interest disclosures were assessed as not having influenced the construction of these Guidelines. Mohammed T Ansari: No conflict of interest to declare. Contracted by SAGES as a methodology consultant. Jason Clay: Nothing to disclose. Rebecca C Dirks: Equity in Johnson & Johnson. Christine Dunst: Nothing to disclose. Geoffrey P Kohn: Nothing to disclose. Lars Lundell: Nothing to disclose. Jeffrey Marks: Consultant fees from Olympus, Boston Scientific. Daniele Molena: Consultant fees from Johnson and Johnson, Urogen, Boston Scientific. Grant from Intuitive. Aurora D Pryor: Speaker for Ethicon, Gore, Merck, Stryker. On scientific advisory board for Obalon. Ceciel Rooker: Nothing to disclose. Payal Saxena: Nothing to disclose. Dimitrios Stefanidis: Nothing to disclose. Lee Swanstrom: Personal fees from Human Xtensions, Titan. Reuben Wong: Nothing to disclose.

Appendix C – Evidence to Decision Tables

The graphical design of the Evidence to Decision Tables is too complicated to render into HTML and the tables are too large to be converted to graphics and rendered in a easy-to-read way. We have therefore extracted the tables as PDF for the best reading experience.

Evidence to Decision Table 1

Evidence to Decision Table 2